UNIT 1 · LESSON 2

The Behavior of Mechanical Waves

How sound travels, spreads out, bounces and changes pitch.

In this lesson you will learn...

- Why mechanical waves need a medium
- Speed, loudness and pitch of sound
- Absorption, transmission and reflection
- The Doppler effect, echolocation and sonar

Words to know

Mechanical wave
A wave that needs a material (a medium) to travel through.

Sound wave
A longitudinal wave made by vibrations.

Intensity / loudness
Intensity: energy flowing through an area. Loudness: how loud it seems to you.

Decibel (dB)
The unit for how intense (loud) a sound is.

Pitch
How high or low a sound seems. Set by the frequency.

Doppler effect
Pitch changes when the sound source moves toward or away from you.

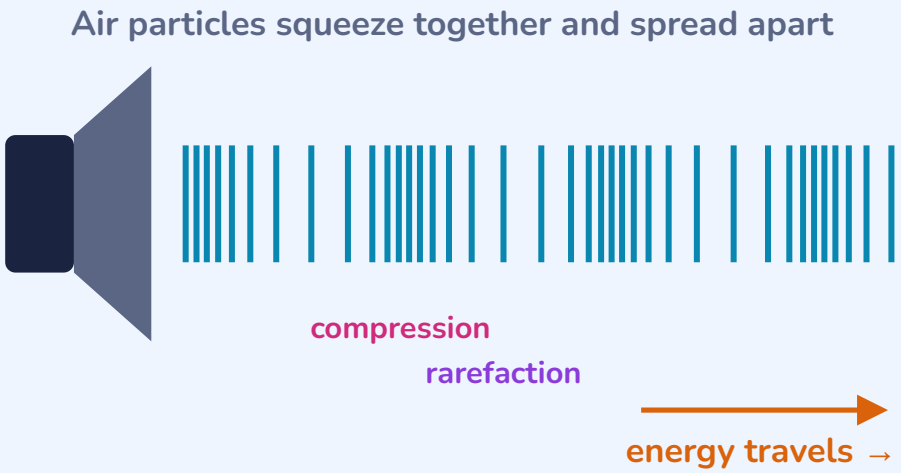
Think about it

A ship floats on the ocean, kilometers above the sea floor. Nobody can see down there.

How can the ship draw a map of the bottom?

Mechanical waves need a medium

- A mechanical wave travels by making the particles of a medium bump into each other.
- Media (plural of medium): air, water, rock, steel…
- No particles = no wave. That's why space is silent.
- When you talk, the air doesn't fly to your friend's ear. The air particles just vibrate and pass the energy along.



How fast is sound?

- Sound travels **faster in solids and liquids** than in gases. The particles are packed tightly and strongly linked, so they pass energy along quicker.
- In air, sound travels **faster when the air is warmer**. Warm air particles move faster.
- Put your ear on a table and tap it: you hear the tap through the wood!

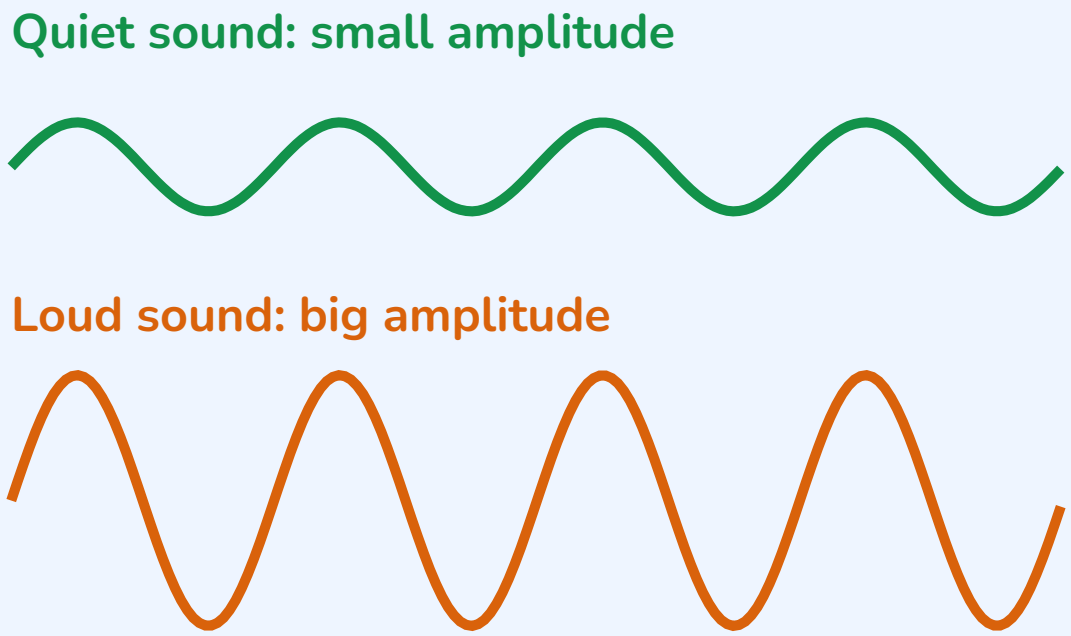
Air
about 340 m/s

Water
about 1,500 m/s

Steel
about 5,900 m/s

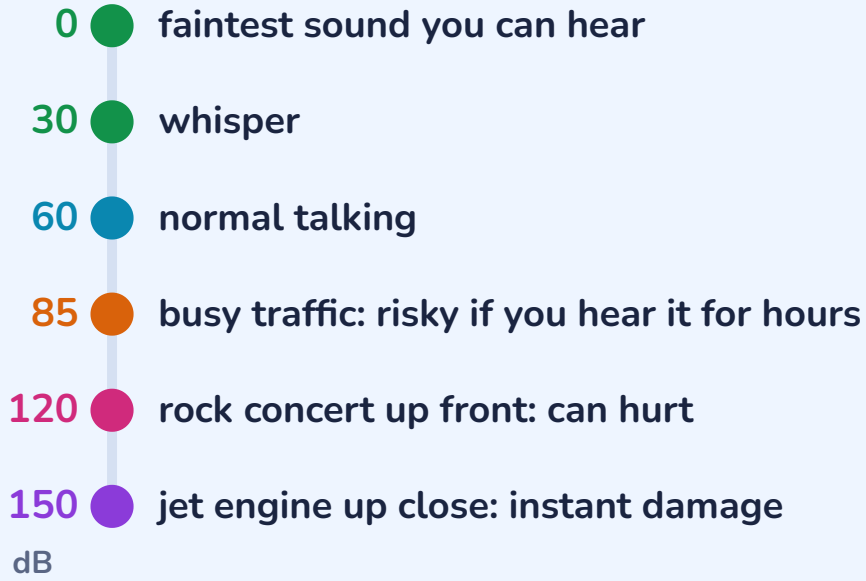
Loudness and pitch

- **Volume ↔ amplitude**: bigger wave = louder sound.
- **Pitch ↔ frequency**: more vibrations per second = higher pitch.
- A mouse squeak: high pitch. A lion's roar: low pitch.



Intensity and decibels

- **Intensity**: how much energy flows through an area each second.
- **Loudness**: how your brain experiences intensity.
- We measure intensity in **decibels (dB)**. 0 dB is the faintest sound a person can hear.
- Very loud sounds (around 120 dB and up) can hurt and damage hearing for good.



Waves spread out

- Mechanical waves spread out in every direction they can.
- The same energy gets spread over a bigger and bigger area.
- So each part of the wave has less energy, and sound gets quieter far from the source.
- Earthquakes shake hardest close to where they start.



Same energy, spread over a bigger circle

Three ways waves meet matter

Absorption

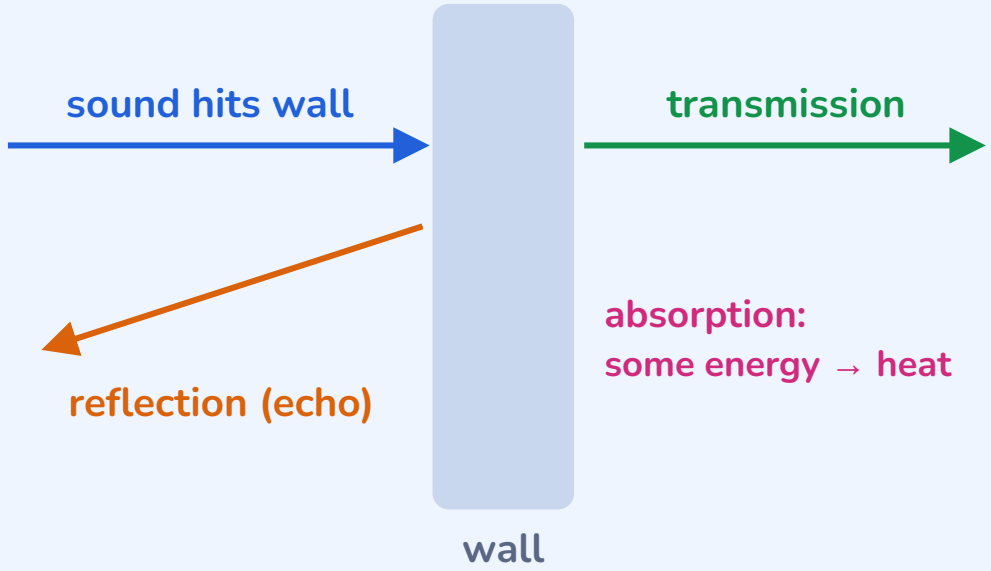
The material soaks up the wave's energy and turns it into heat. High frequencies get absorbed more.

Transmission

The wave passes into and through the material, like hearing music through a wall.

Reflection

The wave bounces back. An echo is reflected sound.



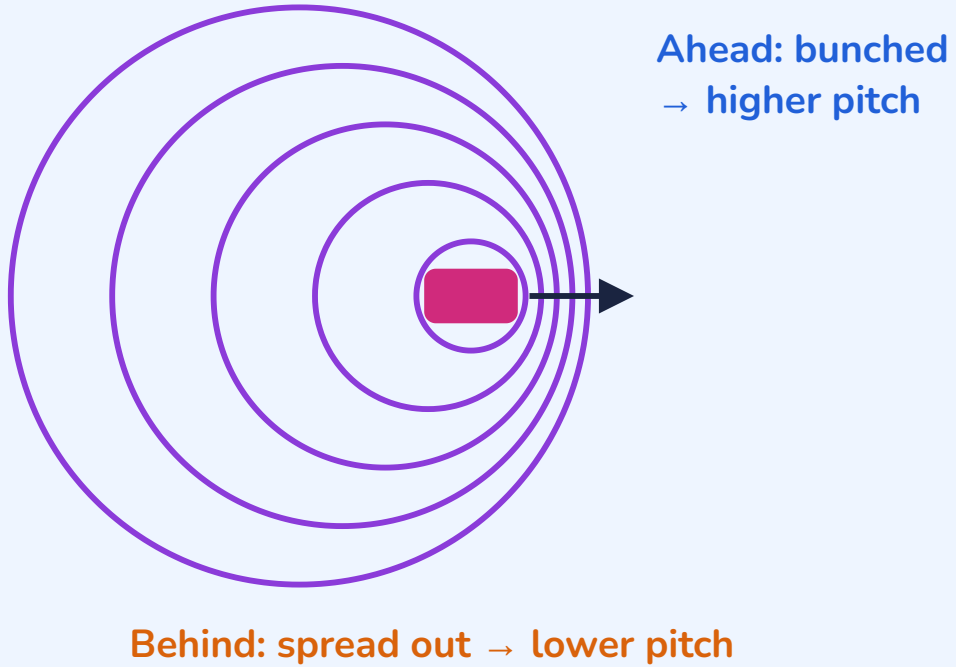
Splitting the energy

- Usually a wave is **partly** reflected and **partly** transmitted.
- One wave becomes two waves.
- The energy is shared, so each new wave is weaker than the original.
- Recording studios use soft materials (foam, carpet, curtains) to absorb sound and cut down echoes.

💡 Look around the room you are in. What absorbs sound? What reflects it? How could you make it quieter?

The Doppler effect

- When a sound source moves **toward** you, the waves bunch up → **higher** pitch.
- When it moves **away**, the waves stretch out → **lower** pitch.
- An ambulance siren goes “WEEEE-wooo” as it passes you.



Using sound to “see”

Echolocation

Send out a sound, listen for the echo. Bats and dolphins find food this way.

Sonar

Ships send sound down through the water and time the echo to map the sea floor.

Ultrasound

Very high-pitched sound makes pictures inside the body, like a baby before it's born.

depth = speed × time ÷ 2

💡 ÷ 2 because the sound goes down **and** back up. Echo after 2 s in water: 1,500 × 2 ÷ 2 = **1,500 m** deep.

Check yourself

1. Why can't astronauts hear each other talk in empty space without radios?
2. Does sound travel faster through a steel rail or through air? Why?
3. Two notes have the same loudness. One has a higher frequency. Which one sounds higher?
4. A sonar echo comes back after 4 seconds. Sound travels at 1,500 m/s in water. How deep is the sea floor?
5. A car honks as it drives away from you. Does the horn sound higher or lower than normal?

Answers

1. Sound is a mechanical wave and needs a medium. Space has almost no particles to carry it.
2. Steel. The particles in a solid are tightly packed and strongly linked, so energy passes faster.
3. The higher-frequency note. Frequency sets pitch.
4. 1,500 × 4 ÷ 2 = 3,000 m.
5. Lower. The sound waves are stretched out as the car moves away.